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import pandas as pd
import random
!pip install faker
from faker import Faker
from datetime import datetime, timedelta

fake = Faker('en_IN')
random.seed(42)

# -----
# CONFIG
# -----
NUM_EMPLOYEES = 100

CAPABILITIES = [
    "AI Native Application Development",
    "Next Gen Application Development",
    "BI & AI Strategies",
    "Client Success"
]

DESIGNATIONS = [
    "Software Engineer",
    "Senior Software Engineer",
    "Tech Lead",
    "Project Manager",
    "Data Analyst",
    "AI Engineer",
    "Consultant",
    "HR Executive",
    "Business Analyst"
]

SHIFT_GROUPS = {
    "General Shift": ("09:00", "18:00"),
    "Morning Shift": ("06:00", "15:00"),
    "Evening Shift": ("14:00", "23:00"),
    "Night Shift": ("22:00", "07:00")
}

WORK_MODES = ["Remote", "Hybrid", "On-site"]

BANKS = [
    "HDFC Bank",
    "ICICI Bank",
    "Axis Bank",
    "State Bank of India",
    "Kotak Mahindra Bank"
]

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]
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```
SKILLS = [  
    "Python", "Power BI", "SQL", "Azure", "AWS",  
    "Machine Learning", "Snowflake", "Java",  
    "React", "Data Engineering", "Leadership",  
    "Communication", "Project Management"  
]
```

```
LEAVE_TYPES = [  
    "Sick Leave",  
    "Casual Leave",  
    "Earned Leave",  
    "Maternity Leave",  
    "Paternity Leave"  
]
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```
EMPLOYMENT_TYPES = ["Full-time", "Contract", "Intern"]
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# HELPERS
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# -----
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```
def random_date(start_year=2024):  
    start = datetime(start_year, 1, 1)  
    end = datetime(2026, 1, 1)  
  
    return start + timedelta(  
        days=random.randint(0, (end - start).days)  
    )
```

```
def indian_mobile():  
    return "+91" + str(random.randint(6000000000, 9999999999))
```

```
def singapore_mobile():  
    return "+65" + str(random.randint(800000000, 999999999))
```

```
def ifsc():  
    return random.choice(["HDFC", "ICIC", "SBIN", "UTIB"]) + "0" +  
    str(random.randint(100000, 999999))
```

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```
# MASTER EMPLOYEE TABLE
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# -----
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employees = []
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manager_ids = []
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```
for i in range(1, NUM_EMPLOYEES + 1):
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```
employee_id = f"EMP{i:03d}"

gender = random.choice(["Male", "Female"])

name = fake.name_male() if gender == "Male" else fake.name_female()

first_name = name.split()[0]
last_name = name.split()[-1]

entity = "Singapore" if i <= 2 else "India"

country = entity

city = random.choice(
    ["Mumbai", "Pune", "Bangalore", "Hyderabad", "Delhi"]
) if entity == "India" else "Singapore"

state = random.choice(
    ["Maharashtra", "Karnataka", "Telangana", "Delhi"]
) if entity == "India" else "Singapore"

phone = singapore_mobile() if entity == "Singapore" else indian_mobile()

alt_phone = singapore_mobile() if entity == "Singapore" else indian_mobile()

dob = fake.date_of_birth(minimum_age=21, maximum_age=55)

age = int((datetime.now().date() - dob).days / 365)

joining_date = random_date(2024)

years_at_company = round(
    (datetime.now() - joining_date).days / 365, 1
)

designation = random.choice(DSIGNATIONS)

employment_type = random.choice(EMPLOYMENT_TYPES)

employment_status = "Active" if i <= 53 else "Exited"

capability = random.choice(CAPABILITIES)

shift_group = random.choice(list(SHIFT_GROUPS.keys()))

shift_start, shift_end = SHIFT_GROUPS[shift_group]
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yearly_salary = random.randint(400000, 3500000)

monthly_salary = round(yearly_salary / 12, 2)

gross_salary = monthly_salary

tax_deduction = round(monthly_salary * 0.12, 2)

pf = round(monthly_salary * 0.05, 2)

other_deduction = round(monthly_salary * 0.03, 2)

net_salary = gross_salary - tax_deduction - pf - other_deduction

rating = round(random.uniform(2.5, 5.0), 1)

goal_completion = random.randint(55, 100)

manager_id = random.choice(manager_ids) if manager_ids else None

if designation in ["Project Manager", "Tech Lead"]:
    manager_ids.append(employee_id)

project_id = f"PRJ{random.randint(1,20):03d}"

employees.append({
    "id": i,
    "employee_id": employee_id,
    "employee_name": name,
    "first_name": first_name,
    "last_name": last_name,

    "personal_email": fake.email(),
    "work_email": f"{first_name.lower()}.{last_name.lower()}@company.com",

    "phone": phone,
    "alternate_phone": alt_phone,

    "dob": dob,
    "age": age,
    "gender": gender,
    "marital_status": random.choice(["Single", "Married"]),
    "nationality": "Indian" if entity == "India" else "Singaporean",
    "blood_group": random.choice(["A+", "B+", "O+", "AB+"]),

    "address": fake.address().replace("\n", " ", " "),
    "city": city,
    "state": state,

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"country": country,
"entity": entity,

"joining_date": joining_date.date(),
"hire_year": joining_date.year,
"years_at_company": years_at_company,
"designation": designation,
"employment_type": employment_type,
"employment_status": employment_status,
"reporting_manager_id": manager_id,

"emergency_contact_name": fake.name(),
"emergency_contact_phone": indian_mobile(),
"emergency_contact_relation": random.choice(
    ["Father", "Mother", "Spouse", "Brother"]
),

"capability": capability,
"allocation_group": random.choice(
    ["Strategic", "Billable", "Internal"]
),
"billable_status": random.choice(
    ["Billable", "Non-Billable"]
),
"project_id": project_id,

"shift_group": shift_group,
"shift_description": shift_group,
"shift_start_time": shift_start,
"shift_end_time": shift_end,
"gross_working_hours": 9,
"break_hours": 1,
"working_days": "Mon-Fri",
"work_mode": random.choice(WORK_MODES),

"bank_name": random.choice(BANKS),
"account_number": str(random.randint(100000000000, 999999999999)),
"account_type": random.choice(["Savings", "Salary"]),
"ifsc_code": ifsc(),

"currency": "SGD" if entity == "Singapore" else "INR",
"yearly_salary": yearly_salary,
"monthly_salary": monthly_salary,
"gross_salary": gross_salary,
"tax_deduction": tax_deduction,
"pf_contribution": pf,
"other_deductions": other_deduction,
"net_salary": net_salary,
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        "rating_score": rating,
        "goal_completion_percent": goal_completion
    })

employee_df = pd.DataFrame(employees)

# -----
# EMPLOYEE SKILLS TABLE
# -----
skills_rows = []

for emp in employees:

    selected_skills = random.sample(SKILLS, random.randint(3, 6))

    for skill in selected_skills:

        skills_rows.append({
            "emp_id": emp["employee_id"],
            "skill_name": skill,
            "skill_category": random.choice(
                ["Technical", "Functional", "Soft Skill"]
            ),
            "proficiency_level": random.choice(
                ["Beginner", "Intermediate", "Advanced", "Expert"]
            ),
            "certification_flag": random.choice(
                ["Yes", "No"]
            )
        })

skills_df = pd.DataFrame(skills_rows)

# -----
# ATTRITION HISTORY
# -----
attrition_rows = []

for idx, emp in enumerate(employees, start=1):

    valid_from = emp["joining_date"]

    valid_to = (
        valid_from + timedelta(days=random.randint(200, 700))
        if emp["employment_status"] == "Exited"
        else None
    )

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attrition_rows.append({
    "hist_id": idx,
    "emp_id": emp["employee_id"],
    "capability": emp["capability"],
    "dept_id": f"DPT{random.randint(1,4):03d}",
    "role_title": emp["designation"],
    "role_id": f"ROL{random.randint(1,20):03d}",
    "manager_id": emp["reporting_manager_id"],
    "employment_type": emp["employment_type"],
    "base_salary": emp["yearly_salary"],
    "currency": emp["currency"],
    "location": emp["city"],
    "country": emp["country"],
    "status": emp["employment_status"],
    "valid_from": valid_from,
    "valid_to": valid_to,
    "change_reason": random.choice([
        "Promotion",
        "Resigned",
        "Department Change",
        "Salary Revision"
    ])
})
})

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attrition_df = pd.DataFrame(attrition_rows)

```

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# -----
# LEAVE DATA
# -----
leave_rows = []

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for i in range(1, 301):

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    emp = random.choice(employees)

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    start_date = random_date(2025)

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    days = random.randint(1, 10)

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    leave_rows.append({
        "leave_id": f"LEV{i:04d}",
        "emp_id": emp["employee_id"],
        "leave_type": random.choice(LEAVE_TYPES),
        "start_date": start_date.date(),
        "end_date": (start_date + timedelta(days=days)).date(),
        "days_taken": days,
        "status": random.choice(

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        ["Approved", "Pending", "Rejected"]
    )
})
```

```
leave_df = pd.DataFrame(leave_rows)
```

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# -----
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```
# EXPORT CSV
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# -----
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```
employee_df.to_csv("employee_master.csv", index=False)
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```
skills_df.to_csv("employee_skills.csv", index=False)
```

```
attrition_df.to_csv("employee_attrition_history.csv", index=False)
```

```
leave_df.to_csv("employee_leave_data.csv", index=False)
```

```
print("CSV files generated successfully!")
```